Yang ZHANG

Ph.D in Mechanical Engineering

14 Avenue de la Paix
92600 Asnières-sur-Seine
France
+33 (0) 609226919
aircraftdesigner66@gmail.com
yang-zhang66
neeyoo

Skills

Programming C/C++, Python, Matlab

Technical
- Software: git, docker, ROS/ROS2, Orocos RTT/eTaSL, Pinocchio, MoveIt!, OpenRMF
- Hardware: Solidworks, Inventor, ABAQUS
- Simulation: Gazebo, Isaac Sim, Visual Component, Simscape, MSC. Adams
- Embedded System: Arduino, STM32, Raspberry Pi
- Real-Time Control System: EtherCAT, Simulink Realtime, Speedgoat, TwinCAT
- Machine Learning: Tensorflow, PyTorch, Scikit-learn
- Computer Vision: Open3D, OpenCV, YOLO
- Robot: Kuka iiwa, UR10, Staubli Tx90, EvoRobot, Mir, Miroki/Miroka

Work Experience

2024.06–now **Senior Robotic Software Engineer**, *Enchanted Tools*, Paris, France.
In the grasping team of Enchanted Tools, I am in charge of developing control algorithms for the dual arm of the the Mirokai robot. I have implemented torque/impedance control, gravity compensation, and QP-based task control on the robot. I also work as a soft / hardware cross-functional role to set up hardware requirements, proof-of-concept design&testing, and testbench design.

2022.10–2024.06 **Research Engineer**, *Flanders Make*, Kortrijk, Belgium.
Working as both researcher and project leader in multiple research projects related to flexible robotic assembly tasks, mobile robot assembly line feeding, skill-based robotic programming, force-based robotic manipulator control for complex tasks. My goal is to simplify robot programming for non-expert users and tackle non-repetitive reconfigurable robotic tasks.

2019.06–2022.10 **Postdoctoral Researcher and Temporary Lecture**, *HEI - JUNIA*, Lille, France.
Participating in the European INTERREG Project - MOTION which has total funding of 7 million euros, taking charge of the design, including mechanical, actuation unit and software, of a lower-limb exoskeleton prototype for children with neurological disorders. At the same time, I am also in charge of teaching Machine Learning and Robotics courses.

Education

2015.11–2019.04 **Ph.D in Mechanical Engineering**, *Institut National des Sciences Appliquées (INSA) de Rennes*, Rennes, France.
Ph.D advisor: Vigen Arakelian
Thesis title: Design and Synthesis of Mechanical Systems with Coupled Units

2012.09–2015.03 **M.Sc in Solid Mechanics**, *Northwestern Polytechnical University*, Xi'an, China.
Master advisor: Erming He
Thesis title:Coupled Dynamic Modeling of Offshore Wind Turbine and its Structural Control
Major courses:Principle of aeroelasticity, Fundamental and application of finite element method, Advanced structural dynamics, Computational fluid dynamics

2008.09– 2012.06 **B.Sc in Aircraft Design and Engineering**, *Northwestern Polytechnical University*, Xi'an, China.
Major courses:Calculus, Probability theory and statistics, Linear algebra, Numerical calculation, Theoretical mechanics, Machinery design, Aerodynamics, Computer programming, Automatic control theory

Research Projects

2022.10– 2024.06 **Symbolic-level robot programming by using skill-based framework and world modeling**

- Design and implement graphical-based world modeling tool for various assembly tasks
- Design of a vendor-independent high-level task orchestrator by implementing skill-based programming framework
- Solving complex robotic tasks by using real-time controller and constrain based methodology (eTaSL)
- CAD design and 3D printing for customized tools and container kit

2023.03– 2024.06 **Flexible assembly infrastructure for heavy and large products**

- Implementing AGV/AMR for logistic taks
- Deploying fleet management system for heterogeneous mobile robots
- Virtual simulation modeling with Nvidia Isaac Sim

2023.10– 2024.06 **Decision support for part feeding of flexible Assembly systems**

- Leading and managing the project within the internal labs
- Model estimating the key factors of resources' process time
- Intralogistics resources operational unitary cost model

2019.06– 2022.10 **Mechanised Orthosis for Children with Neurological Disorders**

- Design of a scalable exoskeleton which can be used by children from 8 to 12 years old.
- Choosing suitable motor and speed reducer for the actuation unit of the exoskeleton through analyzing the gaits of healthy subjects.
- Design of the reference gait and balancing control based on simplified biped walking model (LIP, ICP).
- Developing dynamic model of the exoskeleton for simulation and validating control algorithm.
- Design of a real-time control system by using EtherCAT protocol
- Conducting experimental tests with dummy device and pilots.

2018.01– 2019.01 **Design and Optimization of a Mechanical System Coupling Hand Operated Manipulator with Cobot**

- Design of a coupled system with a collaborative manipulator (UR10) and a hand-operated balanced manipulator for transporting heavy load with a relatively small and cheap robotic arm.
- Analysing the workspace and the singularity of the system.
- Developing the dynamic model of the coupled system for simulation.
- Trajectory optimization for reducing oscillation induced by inertial force.

2016.09– 2018.09 **Design and Synthesis of Gravity balancer for Robotic Manipulator**

- Design of gravity balancers with non-zero length spring and different mechanisms.
- Combining geometric synthesis of the mechanism and conservation of potential energy for optimizing the gravity balancing effect.
- Validation and assessment via computer simulation and experiment.

2015.11– 2017.01 **Design and Synthesis of One Degree-of-Freedom Walking Robot**

- Design of one dof walking robot with different mechanism.
- Developing mathematical representation between the output path of the mechanism and geometrical parameters of the mechanism.
- Optimizing geometrical parameters of the mechanism for generating desired paths.
- Validation of optimized robot via simulation.

Teaching & Mentoring Experience

2020&2021 Oct. - Nov. **Introduction to Machine Learning**, *HEI Junia*, Lille, France.
Description: 25h of theoretical and piratical course including: Linear and logistic regression, Neural network (forward and backward propagation), Error analysis and parameter tuning, K-means, PCA.

2021.10 - 2021.12 **Introduction to Robotics**, *HEI Junia*, Lille, France.
Description: 13h of theoretical course including robot kinematics, dynamics, and position control, 3h of practical course of programming UR5 cobot and design of a controller with Matlab robotic system toolbox to control a 5-dof manipulator in operational space.

2020.09 - 2021.02 **Mentor of Master Student Final-year Project**, *ISEN Junia*, Lille, France.
Description: Mentoring the final-year project of 17 master students in robotics to design an automated unmanned restaurant by using different robotic devices including mobile robots, serial manipulators, automated beverage distributor, etc.

Publications

[1] Yang Zhang, Ali Roshanbin, and Muhammad Raheel Afzal. Graphical world modeling with skill based robot programming for reconfigurable assembly task. In *MECHATRONICS 2023; 17th Mechatronics Forum International Conference*, pages 1–6, 2023.

[2] Ali Roshanbin, Yang Zhang, and Muhammad Raheel Afzal. Self-learning skill parameters for robotic peg-in-hale assembly task. In *2023 11th International Conference on Control, Mechatronics and Automation (ICCMA)*, pages 253–258, 2023.

[3] Yang Zhang, Mathieu Bressel, Sander De Groof, François Dominé, Luc Labey, and Laurent Peyrodie. Design and control of a size-adjustable pediatric lower-limb exoskeleton based on weight shift. *IEEE Access*, 11:6372–6384, 2023.

[4] Yang Zhang and Vigen Arakelian. Design and synthesis of the gravity compensator with a non-zero length spring. *International Journal of Mechanical Engineering and Robotics Research*, 12(3), 2023.

[5] Sander De Groof, Yang Zhang, Laurent Peyrodie, and Luc Labey. Design and control of an individualized hip exoskeleton capable of gait phase synchronized flexion and extension torque assistance. *IEEE Access*, 11:96206–96220, 2023.

[6] Sander De Groof, Yang Zhang, Laurent Peyrodie, Roy Sevit, Emmanuel Vander Poorten, Erwin Aertbelien, and Luc Labey. Design of a series elastic actuator for an assistive exoskeleton using numerical methods and gait data. In *ACTUATOR 2022; International*

Conference and Exhibition on New Actuator Systems and Applications, pages 1–4. VDE, 2022.

[7] Yang Zhang and Vigen Arakelian. Design, optimization and control of a cable-driven robotic suit for load carriage. In *Gravity Compensation in Robotics*, pages 125–146. Springer, 2022.

[8] Yang Zhang and Vigen Arakelian. Design and synthesis of single-actuator walking robots via coupling of linkages. *Frontiers in Mechanical Engineering*, 6:109, 2021.

[9] Yang Zhang, Sander De Groof, Larent Peyrodie, and Luc Labey. Mechanical design of an exoskeleton with joint-aligning mechanism for children with cerebral palsy. In *IEEE BioRob 2020*, Dec 2020.

[10] Yang Zhang and Vigen Arakelian. Legged walking robots: Design concepts and functional particularities. In *Advanced Technologies in Robotics and Intelligent Systems*, pages 13–23, Cham, 2020. Springer International Publishing.

[11] Yang Zhang, Vigen Arakelian, Baptiste Véron, and Damien Chablat. Key features of the coupled hand-operated balanced manipulator and lightweight robot. In *The 15th IFToMM World Congress*, Jun 2019.

[12] Vigen Arakelian and Yang Zhang. An improved design of gravity compensators based on the inverted slider-crank mechanism. *Journal of Mechanisms and Robotics*, 2019.

[13] Vigen Arakelian and Yang Zhang. Improvement of the balancing accuracy in gravity compensators based on the inverted slider crank mechanism. In *ASME 2018 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE 2018)*, Aug 2018.

[14] Yang Zhang and Vigen Arakelian. Design of a single actuator walking robot via mechanism synthesis based on genetic algorithms. *Journal of Robotics and Mechanical Engineering Research*, 2(3):1–7, 2018.

[15] Yang Zhang and Vigen Arakelian. Design of a passive robotic exosuit for carrying heavy loads. In *2018 IEEE-RAS 18th International Conference on Humanoid Robots (Humanoids)*, pages 860–865, Nov 2018.

[16] Yang Zhang, Vigen Arakelian, and Jean-Paul Le Baron. Linkage design for gravity balancing by means of non-zero length springs. In *ROMANSY 22 – Robot Design, Dynamics and Control*, pages 163–170, Jun 2018.

[17] Yang Zhang, Vigen Arakelian, and Jean-Paul Le Baron. Design concepts and functional particularities of wearable walking assist devices and power-assist suits–a review. In *Proceedings of the 58th International Conference of Machine Design Departmants (ICDM'2017)*, pages 436–441, Sep 2017.

[18] Yang Zhang, Vigen Arakelian, and Jean-Paul Le Baron. Design of a legged walking robot with adjustable parameters. In *Proceedings of the 12th International Conference on the Theory of Machines and Mechanisms (TMM 2016)*, pages 65–71, Sep 2016.

References

Laurent Peyrodie, Senior Lecture/Researcher, Department of Smart Energy and System, HEI-Junia, `laurent.peyrodie@junia.com`
Vigen Arakelian, Professor, Department of Mechanical Engineering and Automation, INSA de Rennes, `vigen.arakelyan@insa-rennes.fr`

Languages

- English: C1
- French: B2
- Chinese: Mother Tongue

Interests

- Basketball
- Swimming
- Guitar
- Cycling
- Tennis